

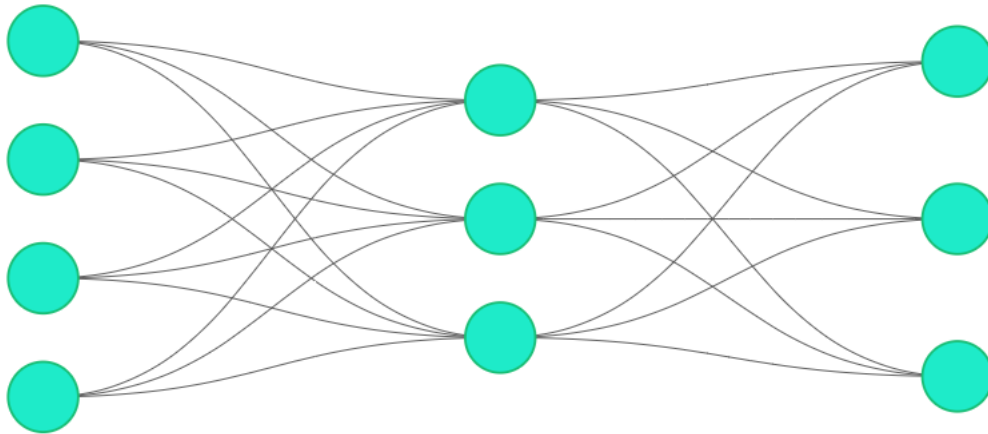
Course : Deep Learning
Week : 3

Information for questions 1 to 8

A team is developing a neural network for *AgroScan*, an AI model used to classify crop conditions into three categories: *Healthy*, *Pest-infected*, or *Nutrient-deficient*. Each plant image is preprocessed to extract 4 numerical features, all normalized between 0 and 1. These features are fed into a feedforward neural network.

The architecture consists of:

- An input layer with 4 neurons,
- A hidden layer with 3 neurons using the sigmoid activation function, and
- An output layer with 3 neurons using softmax activation to produce class probabilities.



The network uses cross-entropy loss during training. All bias terms are zero, and the weight matrices are W_1 (input to hidden) and W_2 (hidden to output).

Question: 1

What is the correct dimension of W_2 in this setup?

Options:

- A. 4×3
- B. 3×4
- C. 3×3
- D. 3×1

Correct Answer: C

Explanation: Hidden has 3 neurons $\rightarrow W_2$ must map (3 hidden $\rightarrow$ 3 output)

Question: 2

Which of the following loss-activation combinations are correctly used in AgroScan?

Options:

- A. Softmax activation and squared error loss
- B. Sigmoid activation and cross-entropy loss
- C. Softmax activation and cross-entropy loss
- D. Linear activation and cross-entropy loss

Correct Answer: C

Explanation:

Softmax activation turns the output into a probability distribution over classes. Cross-entropy measures how well the predicted probabilities match the true class. This combination is standard and appropriate for multi-class classification tasks like AgroScan.

Question: 3

Compute the pre-activation values for the hidden layer for the input $x = [0.2, 0.6, 0.1, 0.5]$

with $w_1 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}$

Options:

- A) [0.9,0.7,0.8]
- B) [0.8, 0.9, 0.7]
- C) [0.8, 0.7, 0.9]
- D) [0.7, 0.9, 0.8]

Correct Answer: C

Explanation :

Pre-activation of the hidden layer is obtained by w_1x

Question: 4

In the AgroScan model, after performing the forward pass for a given input sample, the pre-activation values at the output layer (corresponding to the three classes: Healthy, Pest-infected, and Nutrient-deficient) are: $z = [0.96, -0.27, 1.19]$

Use the softmax function to convert these into probabilities.

Based on the computed probabilities, what class will the network predict for this input?

Options:

- A) Healthy
- B) Pest-infected
- C) Nutrient-deficient
- D) All classes have equal probability

Correct Answer: C

Explanation:

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Question: 5

In a classification task, the true label is “*Pest-infected*“, and the predicted probability for class “*Pest-infected*” is 0.55. Using the cross-entropy loss function, compute the loss.

Fill in the blank with the answer rounded to two decimal places:.

Fill the blank: _____

Answer: allow 0.60 or 0.59

Explanation:

$$L = -\log(0.55) = 0.597837 \approx 0.60$$

Question: 6

Given softmax output [0.35, 0.55, 0.10], what is the predicted class in the *AgroScan* model?

Options:

- A. Healthy
- B. Pest-infected
- C. Nutrient-deficient
- D. None of the above

Answer: B

Question: 7

How many learnable parameters are there in the above neural network in total?

- A. 12
- B. 15
- C. 21
- D. 24

Correct Answer: C

Explanation:

$$w_1 \rightarrow 4 \times 3, w_2 \rightarrow 3 \times 3$$

Question: 8

If the hidden layer used *tanh* instead of *sigmoid*, what would change?

Options:

- A. Activation outputs could be negative
- B. Output layer would become invalid
- C. Pre-activations would change
- D. Number of parameters would increase

Correct Answer: A

Explanation:

tanh can produce both positive and negative outputs (range: -1 to 1)

The output layer uses softmax, which works fine regardless of the hidden activation function.

pre-activations depend on the input and weights, not the activation function itself.

changing the activation function does not affect the number of parameters.

Question: 9

Which of the following can lead to higher cross-entropy loss during training?

Options:

- A. Predicting low probability for the true class
- B. Predicting uniform probabilities for all classes
- C. Predicting 1.0 for the correct class
- D. Predicting a class different from the true one

Correct Answers: A, B, D

Question: 10

Which of the following are always true for softmax output?

Options:

- A. Output values lie in 0, 1
- B. Output values sum to 1
- C. Softmax is invariant to the order of inputs
- D. Softmax is sensitive to relative magnitudes of logits

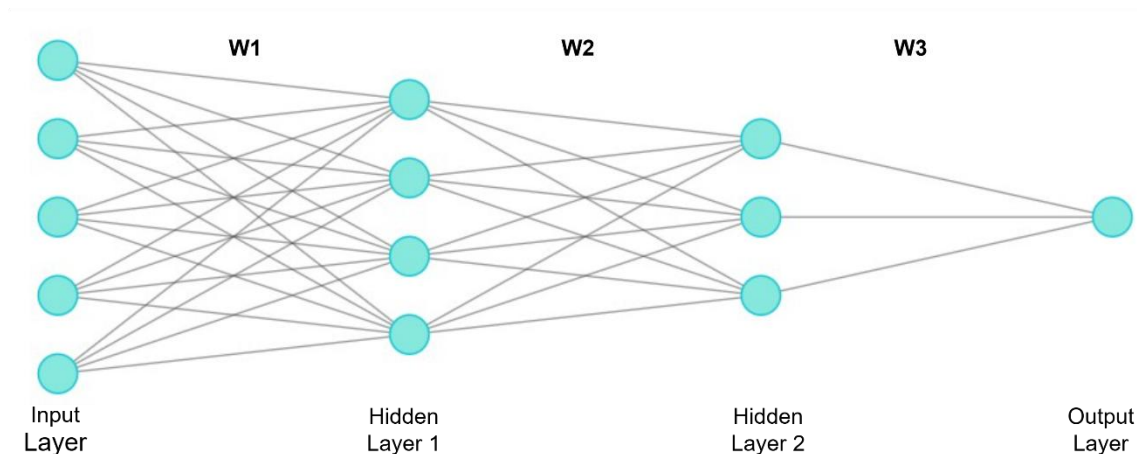
Correct Answers: A, B, D

Case study for the questions from 11 to 20

A research team is developing a neural network model named LeafYield, designed to predict the crop yield (in kg) from leaf-level features extracted via sensors and imaging.

Each sample contains 5 normalized numerical features: Leaf length, Leaf width, Color intensity, Water content, Light absorption level

The neural network contains two hidden layers. First hidden layer has 4 neurons with tanh activation. Second hidden layer has 3 neurons with sigmoid activation. Output layer has one neuron with linear activation.



Question: 11

Why is the output activation in LeafYield chosen to be linear?

Output:

- A. Because softmax is unsuitable for regression
- B. Because we need raw values, not probabilities
- C. Because sigmoid would squash the output range
- D. All of the above

Correct Answer: D

Question: 12

What is the final output if $W_3 = \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$ and activated value of second hidden layer

$$h^{[2]} = \begin{bmatrix} 0.73 \\ 0.90 \\ 0.91 \end{bmatrix} ?$$

Options:

- A) 0.70
- B) 0.71
- C) 0.72

D) 0.73

Correct Answer: C

Question: 13

Which loss function is most appropriate for LeafYield?

Options:

- A. Cross-entropy
- B. Binary cross-entropy
- C. Mean squared error
- D. Kullback–Leibler divergence

Correct Answer: C

Question: 14

Given the number of neurons in the layers and that bias is used in all layers, compute the total number of learnable parameters.

Fill in the blank : _____

Correct Answer: 43

Question: 15

Given output layer activation is linear and final pre-activation value is 3.75, what will be the model's prediction?

Fill in the blank : _____

Correct Answer: 3.75

Question: 16

Which of the following statements are true about the activation functions used in LeafYield?

Options:

- A. tanh outputs values between -1 and 1
- B. sigmoid outputs values between 0 and 1
- C. linear activation has no bounds
- D. sigmoid is more centered around zero than tanh

Correct Answers: A, B, C

Explanation:

tanh is zero-centered; **sigmoid** is not.

Question: 17

Given the input vector

$$x = [0.1, 0.5, 0.3, 0.7, 0.2]^T$$

And $W_1 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$

Which is the pre-activation vector $a^{[1]}$ for hidden layer 1.

Options:

- A) [0.6, 1.2, 0.5, 1.5]
- B) [0.6, 1.2, 0.3, 1.5]
- C) [0.6, 1.3, 0.5, 1.5]
- D) [0.6, 1.3, 0.3, 1.5]

Correct Answer: B

Question:18

Apply tanh activation to the above vector obtained in Question: 17 and round to two decimals.

Options:

- A) [0.54, 0.83, 0.46, 0.91]
- B) [0.54, 0.83, 0.29, 0.91]
- C) [0.54, 0.86, 0.46, 0.91]
- D) [0.54, 0.86, 0.29, 0.91]

Correct Answer: B

Question: 19

The LeafYield model was tested on 5 plant samples. The true and predicted crop yields (in kg) are:

Sample	True Yield	Predicted Yield
1	3.0	2.7
2	4.5	4.2
3	2.0	2.5
4	3.5	3.0
5	5.0	5.3

What is the Mean Squared Error?

Options:

- A) 0.12
- B) 0.15
- C) 0.25

D) 0.30

Correct Answer: B

Explanation:

MSE is the average of squared differences.

$$MSE = \frac{(0.3)^2 + (0.3)^2 + (0.5)^2 + (0.5)^2 + (0.3)^2}{5} = 0.154 \approx 0.15$$

Question: 20

Which layer's weights will receive gradients first during backpropagation?

Options:

- A. Output layer
- B. Hidden layer 2
- C. Hidden layer 1
- D. Input layer

Correct Answer: A

Explanation: Backpropagation flows from output to input.

Case study for Questions 21 to 30

A sports analytics company has developed *PlayPredict*, a neural network model that classifies football players into one of four roles based on physical and tactical attributes, such as *Defender*, *Midfielder*, *Forward*, *Goalkeeper*.

From video and tracking data, 5 numerical features are extracted: Speed, Pass accuracy, Defensive actions, Dribble attempts, Position heatmap score.

The model has one hidden layer has 4 neurons with tanh activation and an output layer.

Question 21:

Which activation is best suited for the above scenario?

Options:

- A) Linear activation
- B) Sigmoid activation
- C) Tanh activation
- D) Softmax activation

Correct Answer: D

Question: 22

What is the predicted class if the output vector contains [0.2, 0.1, 0.6, 0.1]?

Options:

- A) Defender
- B) Midfielder
- C) Forward
- D) Goalkeeper

Correct Answer : C

Explanation:

Class with highest softmax score

Question: 23

Given logits: [1.2, 0.8, 2.0, 1.0], compute softmax output (rounded to 2 decimals)

Options:

- A) [0.22, 0.14, 0.48, 0.17]
- B) [0.21, 0.14, 0.47, 0.17]
- C) [0.28, 0.16, 0.58, 0.27]
- D) [0.51, 0.54, 0.47, 0.17]

Correct Answer: B

Question: 24

Why use tanh in the hidden layer?

Options:

- A. Keeps outputs in [0,1]
- B. Introduces non-linearity
- C. Allows negative outputs
- D. Helps output softmax

Correct Answer: B, C

Explanation:

tanh is non-linear and zero-centered

Question: 25

Given the input feature vector $x = [0.5, 0.3, 0.2, 0.1, 0.4]^T$ and

weight matrix $W_1 = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$.

Compute the pre-activation vector $a^{[1]}$.

Options:

- A. [0.9, 0.9, 0.8, 0.7]
- B. [0.9, 0.8, 1.1, 0.6]
- C. [0.9, 0.9, 1.1, 0.7]
- D. [1.0, 0.9, 1.1, 0.8]

Correct Answer: A

Question: 26

Compute the hidden vector $h^{[1]}$ from the above computed $a^{[1]}$.

Options:

- A. [0.72, 0.72, 0.66, 0.60]
- B. [0.72, 0.66, 0.80, 0.54]
- C. [0.72, 0.72, 0.80, 0.60]
- D. [0.76, 0.72, 0.80, 0.66]

Correct Answer: A

Question: 27

The PlayPredict model produced the following softmax output for a football player sample:

$$\hat{y} = [0.1, 0.7, 0.1, 0.1]$$

What is the categorical cross-entropy loss for this prediction (rounded to 3 decimals)?

Options:

- A. 0.105
- B. 0.357
- C. 0.500
- D. 0.845

Correct Answer: B

Explanation:

The correct class label for this player is class 2 (Midfielder).

For cross-entropy, the loss is: $-\log(0.7) \approx 0.357$

Question: 28

When bias is used in both the layers, what is the total number of learnable parameters in the playpreidct network?

Fill in the blank _____

Correct Answer: 44

Explanation:

$$(5 \times 4) + 4 + (4 \times 4) + 4 = 44$$

Question: 29

Which of the following can be tanh activation output?

Options:

A. $[-0.9, 0.2, 0.8, -0.4]$

B. $[1.5, -1.2, 2.3, 0.5]$

C. $[0.0, 1.0, 2.0, -1.0]$

D. $[-2, -1, 0, 1]$

Correct Answer: A

Explanation: tanh outputs are between -1 and 1 .